

AKITA, Japan

WMO: 475820

Lat: 39.7172N

Lon: 140.0994E

Elev: 22

StdP: 101.06

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.8	-3.9	-10.5	1.5	-1.4	-9.4	1.7	-1.0	14.3	-0.2	13.2	-0.4	5.3	310	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	31.9	24.5	30.4	23.9	28.9	23.3	25.5	29.5	24.9	28.9	24.3	27.9	4.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.5	19.5	27.6	23.7	18.6	27.2	23.1	17.9	26.8	78.7	29.5	76.2	29.1	73.6	28.0	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.5	9.2	-6.9	34.7	1.2	1.6	-7.8	35.9	-8.5	36.8	-9.2	37.7	-10.1	38.9		
(5)			-7.6	26.3	1.2	0.8	-8.5	26.9	-9.2	27.3	-9.9	27.7	-10.7	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	0.3	0.8	4.0	9.5	15.4	19.7	23.8	25.4	21.2	14.7	8.3	2.9
	DBStd	9.15	2.25	2.59	2.91	3.36	3.09	2.50	2.46	2.42	3.12	2.94	3.40	2.78
	HDD10.0	1091	301	258	188	49	1	0	0	0	0	2	72	221
	HDD18.3	2779	559	491	445	265	101	14	0	0	8	116	302	479
	CDD10.0	1902	0	0	1	34	168	291	427	477	335	149	21	1
	CDD18.3	549	0	0	0	0	9	55	169	218	93	5	0	0
	CDH23.3	3713	0	0	0	2	55	216	1085	1842	505	8	0	0
	CDH26.7	991	0	0	0	0	6	33	281	558	112	1	0	0
Wind	WSAvg	4.3	5.1	4.9	4.9	4.6	4.1	3.6	3.5	3.6	3.6	3.9	4.5	5.1
Precipitation	PrecAvg	1749	126	96	100	124	122	119	190	186	172	161	184	166
	PrecMax	2375	213	209	185	399	289	355	543	453	369	288	365	311
	PrecMin	1235	51	50	48	20	36	19	15	16	53	35	66	78
	PrecStd	251	38	33	31	53	49	66	99	93	72	53	63	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	10.7	15.7	22.6	27.3	29.9	33.8	34.4	32.3	24.3	19.0	13.3
		MCWB	6.0	6.9	8.8	13.3	16.7	20.8	24.8	25.0	24.3	18.4	14.6	11.0
	2%	DB	6.3	8.1	12.6	19.6	24.8	27.3	31.2	32.3	29.5	22.5	17.0	10.7
		MCWB	3.8	5.1	8.2	12.2	16.2	19.5	24.2	24.8	23.3	17.3	13.2	8.2
	5%	DB	4.8	6.4	10.7	17.4	22.9	25.8	29.5	31.0	27.6	21.2	15.3	9.0
		MCWB	2.5	3.6	7.5	11.0	15.4	19.1	23.4	24.4	22.2	16.5	12.1	6.7
	10%	DB	3.7	5.0	9.1	15.2	20.9	24.2	28.0	29.7	26.1	19.9	13.9	7.3
		MCWB	1.4	2.4	6.3	10.4	14.9	18.5	22.9	24.1	21.3	15.6	11.0	4.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	8.0	10.5	15.1	18.7	22.3	25.8	26.5	25.2	20.5	15.8	11.4
		MCDB	7.9	9.6	13.3	19.5	23.4	25.3	30.2	30.7	29.6	22.9	17.8	12.7
	2%	WB	4.3	5.7	9.2	13.3	17.6	21.2	25.1	25.9	24.1	18.7	14.2	8.5
		MCDB	5.8	7.6	11.6	17.4	21.8	24.3	29.3	29.7	28.0	21.1	16.0	10.3
	5%	WB	2.9	4.0	7.9	12.2	16.6	20.5	24.4	25.3	23.1	17.4	12.7	6.8
		MCDB	4.3	5.9	10.3	15.6	20.7	23.6	28.2	29.2	26.9	20.1	14.8	8.7
	10%	WB	1.8	2.5	6.5	11.2	15.7	19.7	23.8	24.8	22.2	16.2	11.3	5.2
		MCDB	3.3	4.6	9.0	14.5	19.7	22.9	27.1	28.7	25.5	19.0	13.4	7.1
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.4	6.8	8.1	7.9	7.0	6.1	6.8	7.5	8.0	7.0	5.2
		MCDBR	6.4	8.0	9.8	11.9	11.7	9.9	8.8	8.6	8.6	9.3	9.3	8.0
		MCWBR	5.1	6.2	6.9	6.4	5.5	4.5	3.8	3.7	4.1	5.5	6.9	6.8
	5% WB	MCDBR	6.4	7.6	8.8	8.9	8.7	7.1	6.8	6.9	7.2	7.3	8.2	7.7
		MCWBR	5.7	6.7	7.2	5.9	5.0	3.9	3.2	3.4	4.0	5.5	7.2	7.1
Clear-Sky Solar Irradiance	taub	0.399	0.438	0.508	0.549	0.561	0.537	0.546	0.513	0.455	0.430	0.422	0.408	
	taud	1.980	1.896	1.812	1.789	1.829	1.970	2.012	2.115	2.233	2.172	2.091	2.018	
	Ebn at Noon	714	745	736	740	742	761	749	764	786	755	691	668	
	Edh at Noon	128	160	194	212	208	181	172	151	127	121	115	115	
All-Sky Solar Radiation	RadAvg	1.29	2.07	3.23	4.45	4.99	5.08	4.47	4.62	3.81	2.84	1.71	1.14	
	RadStd	0.10	0.25	0.29	0.45	0.52	0.46	0.54	0.52	0.28	0.21	0.17	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.37	N/A	N/A	N/A	+69
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page